2. Use an interpreter pattern to implement the addition and subtraction rules of single digits. Write a Test class to calculate the expression “5-2+9”.

AdditionExpression

```
package experiment23_2;


class AdditionExpression implements Expression{
    private Expression leftExpression;
    private Expression rightExpression;
    public AdditionExpression(Expression leftExpression,
Expression rightExpression){
        this.leftExpression = leftExpression;
        this.rightExpression = rightExpression;
    }

    @Override
    public int interpret() {
        return leftExpression.interpret() +
rightExpression.interpret();
    }
}
```

Expression

```
package experiment23_2;


interface Expression {
```

```
    int interpret();
}
```

Number

```
package experiment23_2;

class Number implements Expression{
    private int num;
    public Number(int num){
        this.num = num;
    }

    @Override
    public int interpret() {
        return num;
    }
}
```

SubtractionExpression

```
package experiment23_2;

class SubtractionExpression implements Expression{
    private Expression leftExpression;
    private Expression rightExpression;
    public SubtractionExpression(Expression leftExpression,
```

```
Expression rightExpression){
        this.leftExpression = leftExpression;
        this.rightExpression = rightExpression;
    }

    @Override
    public int interpret() {
        return leftExpression.interpret() -
rightExpression.interpret();
    }
}
```

Test

```
package experiment23_2;

public class Test {
    public static void main(String[] args) {
        Expression five = new Number(5);
        Expression two = new Number(2);
        Expression nine = new Number(9);

        Expression subtraction = new
SubtractionExpression(five, two);
```

```
        Expression addition = new
AdditionExpression(subtraction, nine);
        System.out.println("Expression Result (5 - 2 + 9): " +
addition.interpret());


    }
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\
Expression Result (5 - 2 + 9): 12

Process finished with exit code 0
```